

Part 3: Alpha Signal & Validation — Results (Corrected Alignment)

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Summary

After correcting the signal timing, the leader's **common** move predicts next-day follower returns, but the leader-**specific** shock does not, so the composite conditional signal fails hypothesis 3.

- **H1 (continuation on the common component): supported.** `common_lag` $h=1$ rank IC = 0.0049 ($t = 3.52$); quintile long-short = 7.1 bps/day ($t = 6.76$).
- **H2 (reversion on the leader-specific shock): not supported.** `shock_lag` $h=1$ IC = 0.0007 ($t = 0.72$), with the wrong sign for reversion; long-short $t = 0.76$.
- **H3 (conditioning beats undifferentiated reversal): rejected.** The composite's $h=1$ IC (0.0010, $t = 0.93$) is below both baselines: the Part 1 leader-return signal (0.0021) and naive own-return reversal (0.0166).

The composite fails because the shock leg adds noise to a working common leg, not because the leader carries no information.

Correction to the preliminary results

The preliminary Part 3 results (Sep 25, 2026) tested the signal one day too late; all Part 3 numbers in this version replace them.

- **The bug:** `build_conditional_panel(lag=1)` puts the leader's move from day $t-1$ on row t . `horizon_ic` then scores row t against `fwd_ret_1d`, which is the return on day $t+1$ alone. The effective gap was **two days**, not one.
- **The fix:** `run_part3.py` now calls `conditional_signal_panel(..., lag=0)`. Row t carries the leader's day- t move, known at day t 's close, tested against the day $t+1$ return: a one-day gap with no look-ahead, since the shocks come from the point-in-time `decompose_rolling`.
- **Unchanged:** Parts 1 and 2. Their Fama-MacBeth regressions put the same row's return on the left-hand side, so `lag=1` is correctly a one-day gap there. The default `lag=1` in `build_conditional_panel` is kept for that reason.

- **Fair comparison preserved:** the Part 1 baseline (`leader_ret_lag`) and naive reversal (`own_lag`) come from the same panel, so they shifted with the fix.

Signal and hypotheses

The conditional signal turns Part 2's split of each leader's move into one rankable number per follower-day.

$$\text{conditional_signal} = \text{common_lag} - \text{shock_lag}$$

- **H1:** followers continue on the leader's common component, so `common_lag` enters with a plus.
- **H2:** followers revert on the leader-specific shock, so `shock_lag` enters with a minus.
- **H3:** conditioning on this split beats both the unconditional leader signal (Part 1) and naive own-return reversal.

Each leg is also tested on its own. A composite that works while one leg is flat is a weaker finding than both legs working.

Methodology

Every signal is scored the same way, on the same follower panel, with no look-ahead.

Item	Setting
Follower panel	12,388,705 follower-days, 7,079 stocks (Part 1 output)
Leader decomposition	<code>decompose_rolling</code> : point-in-time, 298,562 industry-days
Signal coverage	12,245,966 follower-days (98.8%) with a non-missing composite
Timing	Signal known at day t close; tested on returns from $t+1$
Horizons	1, 5, 10, 20 trading days (compounded forward returns)

Item	Setting
Predictive power	Daily cross-sectional Spearman rank IC, averaged over about 7,100 days
Economic size	Daily quintile sort; long Q5, short Q1, at $h = 1$

The pre-registered headline is $h = 1$. Scanning other horizons or subsamples for a better result would be p-hacking, as noted in `docs/STATUS.md`.

Results

At the headline one-day horizon, the common leg is the strongest leader-based signal and the composite is the weakest.

Headline, $h = 1$

Signal	Mean IC	IC t-stat	Long-short (bps/day)	Long-short t-stat	Days
<code>conditional_signal</code> (common – shock)	0.0010	0.93	3.0	3.72	7,113
<code>common_lag</code> (continuation leg)	0.0049	3.52	7.1	6.76	7,113
<code>shock_lag</code> (reversion leg, raw sign)	0.0007	0.72	0.5	0.76	7,113
<code>leader_ret_lag</code> (Part 1 baseline)	0.0021	1.87	3.4	3.91	7,111
<code>reversal_signal</code> (naive, –own_lag)	0.0166	10.91	3.8	3.23	7,173

IC t-statistics at longer horizons

Signal	$h = 5$	$h = 10$	$h = 20$
<code>conditional_signal</code>	0.19	–0.59	1.39
<code>common_lag</code>	1.65	2.01	3.34
<code>shock_lag</code>	1.27	1.69	0.10
<code>leader_ret_lag</code>	1.28	1.33	1.41
<code>reversal_signal</code>	11.62	9.75	7.71

Longer-horizon t-statistics use overlapping forward returns and may be overstated (see Caveats). Full tables: `results/p3_horizon_ic.csv`, `results/p3_quintile_spread_h1.csv`.

Interpretation

The leader's information that reaches followers is the part shared with the industry, not the part unique to the leader.

- **Common leg works.** Quintile returns rise almost monotonically, from 1.3 bps/day in Q1 to 8.4 bps/day in Q5. Splitting the leader's move doubles the long-short spread of the undifferentiated Part 1 signal (7.1 vs 3.4 bps/day).
- **Shock leg is flat.** Quintile returns show no pattern (4.6 to 5.8 bps/day), and the IC sign is opposite to H2. There is no evidence that followers overreact to leader-specific news and then revert.
- **Composite is diluted.** Subtracting a leg with no predictive power cuts the common leg's IC by about 80% (0.0049 to 0.0010). The composite's long-short (3.0 bps/day) is below the Part 1 baseline.
- **Naive reversal leads on IC but not on spread.** Its IC t-stat (10.9) is the largest, yet its long-short (3.8 bps/day) is about half the common leg's. Its Q5 is the most volatile quintile (1.75% daily std vs about 1.3% elsewhere), consistent with small, illiquid stocks and bid-ask bounce.

A stronger Part 3 signal is `common_lag` alone. Because it was not the pre-registered composite, it should be reported as a finding, not substituted as the headline.

Caveats and next steps

- ☐ **Overlapping horizons.** At $h = 5, 10, 20$, consecutive forward returns overlap, so daily ICs are autocorrelated. Confirm whether `compute_ic` uses Newey-West; if not, the longer-horizon t-stats are overstated.
- ☐ **Sample mismatch.** Naive reversal covers 7,173 days vs 7,113 for the leader signals. Rerun the comparison on the common set of days.
- ☐ **Bid-ask bounce.** Check the reversal baseline against `scripts/bounce_check.py`, or skip a day, before treating it as tradable.
- ☐ **Transaction costs.** Spreads of 3 to 7 bps/day before costs need a turnover and cost estimate before any trading claim.

- ☐ **Downstream use.** Part 4 should build on the corrected `lag=0` signal from the current repo.